

Mostafa Tamer Mostafa Mohammed

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Professional Summary

Owned the end-to-end architecture of the critical ECG device-to-server communication system at Pulse, driving production stability that directly secured enterprise contract renewals in the UK and new regional expansions across major healthcare networks in Egypt and Saudi Arabia. **Focused on solving complex concurrency, real-time data ingestion, and network protocol failures in production environments.**, with a strong foundation in modular Clean Architecture and Microservices concepts.

Work Experience

Java Software Engineer (Backend) – **Pulse for Integrated Solutions GmbH** (Dec 2024 – Present)

- **High-Availability Streaming Pipelines:** Engineered an atomic, multi-stage request pipeline for critical ECG live-streaming operations, ensuring zero stream drops in hospital environments via dynamic stage-level retries; eliminated catastrophic stream failures and protected system runtime by auto-aborting orphaned pipelines upon sudden device disconnections.
- **Device Manager Architecture:** Architected and developed a centralized module to manage and control ECG medical devices, refactoring legacy business logic out of a monolithic communication class into a scalable, modular design.
- **Connection Concurrency Control:** Resolved a critical race condition where network instability caused duplicate active connections for a single device, implementing an eviction strategy that forcibly terminates stale connections to prevent state corruption and database inconsistencies.
- **Data Integrity & Crash Recovery:** Engineered a thread-safe Backup Manager to synchronize device SD card data to the server; built a robust **server-boot recovery routine** that scans backup files to locate the last valid frame and isolate physical corruption.
- **Critical Incident Resolution (EgyptAir POC):** Partnered with the mobile engineering team to diagnose and resolve a high-priority Bluetooth streaming defect caused by frame serial inconsistencies and corrupted delay calculations; stabilized the system within a critical window, successfully securing a major enterprise contract with EgyptAir.
- **Protocol Integration & Ownership:** Acted as the primary backend point of contact for the embedded team to co-design and implement the communication protocol for core device features; integrated multi-master "busy detection" and adaptive network timeouts to stabilize session lifecycles, real-time streaming, and device backups.
- **Hardware Abstraction & Safety:** Developed an **Adapter layer** to isolate core business logic from **different hardware versions** via a unified interface; utilized Java switch expressions to enforce compile-time safety across all hardware iterations.
- **Memory Optimization & Profiling:** Diagnosed and eliminated memory leaks by analyzing heap dumps and using IntelliJ Profiler, resolving root causes linked to static map retention and thread race conditions within executor service lifecycles.
- **Self-Healing & Observability Infrastructure:** Developed a dual-layer device health monitor that automatically triggers silent recovery mechanisms for production failures, while exposing real-time diagnostic flags during testing to instantly pinpoint hardware-side failures.
- **Integration Testing & Architecture:** Introduced the first automated integration testing framework for the system using Arquillian, overcoming physical hardware integration challenges to validate core session and real-time streaming states under complex network and device failure scenarios.

Prior Android Experience

Android Developer | Manzoma Technology Solutions (2023 – 2024)

Developed and optimized large-scale enterprise applications (Assets Manager, Inventory Manager, and CMMS) utilizing Kotlin, XML, and Jetpack Compose.

Technical Skills

- **Languages & Core Foundations:** Java, Kotlin, C/C++, SQL, Concurrency & Multithreading, OOP, Data Structures & Algorithms, SOLID Principles, Design Patterns.
 - **Backend & IoT Integration:** Spring Boot, Jakarta EE, MQTT Protocols, Real-Time Data Ingestion, WebSockets (Architecture & Channel Isolation), REST APIs, Stream Processing, Dependency Injection, Maven, Gradle.
 - **Frontend & Web Development:** HTML5, CSS3, JavaScript (ES6+), Tailwind CSS, Bootstrap, Angular (Familiar).
 - **Databases & Performance Tuning:** MySQL, PostgreSQL, Query Optimization, Indexing Strategies (B-Tree, Covering, Composite), Stored Procedures & Triggers, Concurrency Control.
 - **Architecture & Systems Design:** Microservices Architecture, Clean Architecture, Domain-Driven Design (DDD), Resilience Patterns, Service Decomposition, Eventual Consistency.
 - **DevOps & Cloud Infrastructure:** Docker, Docker Compose, Kubernetes Fundamentals, AWS (EC2), Linux.
 - **Tools & Testing:** Git/GitHub, IntelliJ Profiler, Arquillian Integration Testing, Postman.
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Side Projects

Large-Scale E-Commerce Database Design

- Designed and populated a massive PostgreSQL database, engineering custom functions to efficiently handle millions of rows. Optimized complex SQL queries and index strategies, resulting in a **74.7% performance improvement**.
- [GitHub](#)

Real-Time Chat Application

- Engineered a full-stack Android and Spring Boot chat system supporting real-time interactions via WebSockets and REST APIs. Integrated Firebase Cloud Messaging (FCM) for background push notifications and managed data models for user relationships and group chats.
 - [Backend GitHub](#) | [Android GitHub](#)
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Education & Certifications

- **B.Sc. in Computer Science** – Cairo University, FCAI (2024)
- **Cumulative Grade:** Very Good.
- **Graduation Project:** "Task Together" (A+ Grade).
- **Advanced Android Kotlin Development** – egFWD Scholarship (Udacity, 2022).